



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

CHEMISTRY
PAPER 1

9189/1

JUNE 2014 SESSION

2 hours

Additional materials:

Answer paper Graph paper

Data Booklet

Mathematical tables and/or electronic calculator

TIME: 2 hours

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces provided on the answer paper/answer booklet.

Answer **six** questions.

Answer **two** questions from Section A, **one** question from Section B, **two** questions from Section C and **one** other question chosen from any section.

Write your answers on the separate answer paper provided.

If you use more than one sheet of paper, fasten the sheets together.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

You are reminded of the need for good English and clear presentation in your answers.

This question paper consists of 8 printed pages.

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Section A

Answer at least two questions from this section.

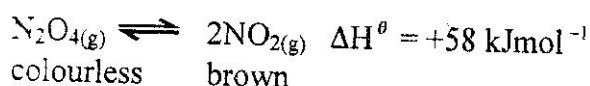
- 1 (a) (i) Write the electronic configuration of
1. calcium,
 2. chlorine.
- (ii) Draw a dot-and-cross diagram to illustrate the type of bond formed between calcium and chlorine. [3]
- (b) (i) Define the term *enthalpy change of hydration*.
- (ii) Write an equation for the enthalpy change of hydration of the
1. calcium ion,
 2. chloride ion.
- (iii) Given that:
- lattice energy of calcium chloride = $-2\,258\text{ kJmol}^{-1}$
enthalpy change of hydration of calcium ions = $-1\,650\text{ kJmol}^{-1}$
enthalpy change of solution of calcium chloride = -120 kJmol^{-1}
1. construct an energy cycle for these enthalpies.
 2. use the cycle to calculate the enthalpy change of hydration of the chloride ion. [6]
- (c) The enthalpy change of hydration of magnesium ion is $-1\,921\text{ kJmol}^{-1}$.
- (i) Use your value in (b)(iii) to calculate the total enthalpy change of hydration of the ions in magnesium chloride.
- (ii) Comment on the difference between the enthalpy of hydration of the magnesium and calcium ions.

[3]

[Total: 12]

- 2 (a) With the aid of a labelled diagram, describe the electrolysis of brine using a diaphragm cell. [7]
- (b) By reference to relevant E^θ values from the Data Booklet,
- (i) explain the disadvantage of using a silver anode,
- (ii) justify why electricity must be supplied for the electrolysis of brine. [5]
- [Total: 12]

- 3 (a) The following equation represents an equilibrium reaction:



- (i) State LeChatelier's principle.
- (ii) Describe and explain the colour change(s) if any, that might take place when
1. a mixture of $\text{NO}_{2(g)}$ and $\text{N}_2\text{O}_{4(g)}$ is compressed at constant temperature,
 2. a mixture of $\text{NO}_{2(g)}$ and $\text{N}_2\text{O}_{4(g)}$ is heated at constant pressure.
- [5]
- (b) In an experiment to determine the equilibrium constant for the reaction in (a) at 200°C , the initial concentrations of N_2O_4 and NO_2 were $0.075 \text{ mol dm}^{-3}$ and $0.0250 \text{ mol dm}^{-3}$ respectively. If the equilibrium concentration of N_2O_4 was $0.00275 \text{ mol dm}^{-3}$,
- (i) calculate the equilibrium concentration of NO_2 ,
- (ii) determine the K_c value of the $\text{N}_2\text{O}_4/\text{NO}_2$ system at 200°C . [4]
- (c) NO_2 can catalyse the conversion of atmospheric sulphur dioxide to sulphur trioxide.
- (i) Write equations to illustrate this conversion.
- (ii) Explain the type of catalysis exhibited by NO_2 in this conversion. [3]
- [Total: 12]

Section B

Answer at least **one** question from this section.

- 4 (a) The boiling point of lead (IV) chloride cannot be measured directly by heating. **Table 1** shows the boiling points of three Group (IV) chlorides.

Table 1

chloride	boiling point/ $^{\circ}\text{C}$
SiCl_4	58
GeCl_4	83
SnCl_4	114

- (i) Plot a line graph of mass number against boiling point.
- (ii) Extrapolate your graph to deduce the boiling point of PbCl_4 .
- (iii) Suggest a reason why the boiling point of PbCl_4 cannot be measured directly by heating.
- (iv) Explain the trend in boiling points of the tetrachlorides.

[6]

- (b) Silicon tetrachloride can be reduced to silicon by heating with zinc granules.

- (i) Write a chemical equation for the reaction.
- (ii) A sample of silicon tetrachloride contained 12% by mass of impurities.

Calculate the mass of silicon that could be obtained by heating 125 g of the impure tetrachloride with an excess of zinc.

[3]

- (c) Trilead tetraoxide contains lead atoms in two different oxidation states.

- (i) Write the formula of trilead tetraoxide.
- (ii) Suggest the oxidation states of lead in trilead tetraoxide.
- (iii) Write the chemical equation for the action of heat on trilead tetraoxide.

[3]

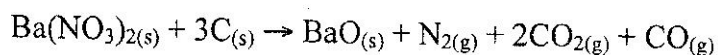
[Total: 12]

- 5 (a) Table 2 shows the decomposition temperatures of Group (II) carbonates.

Table 2

carbonate	MgCO ₃	CaCO ₃	SrCO ₃	BaCO ₃
decomposition temperature/°C	540	900	1 290	1 360

- (i) Write the chemical equation for the decomposition of SrCO₃.
- (ii) Account for the trend in the decomposition of the Group (II) carbonates. [4]
- (b) An equal mass of each of the Group (II) oxides was placed in a test tube and excess sulphuric acid added. Each test tube was then thoroughly shaken and allowed to stand.
- (i) Describe and explain the appearance of the products, after the reaction in the test tube containing,
- magnesium oxide,
 - calcium oxide.
- (ii) Write the chemical equation for the reaction of magnesium oxide with sulphuric acid. [5]
- (c) The equation for the reaction between barium nitrate and carbon when heated is as follows:



Calculate the total volume of the gas given off, at r.t.p. when a 10.0 g sample of the mixture of barium nitrate and carbon is heated. [3]

[Total: 12]

Section C

Answer at least **two** questions from this section.

- 6 (a) Methylpropene, $(\text{CH}_3)_2\text{CCH}_2$, reacts with hydrogen bromide.
- Name, with a reason, the type of reaction undergone by methylpropene and hydrogen bromide.
 - Describe the mechanism for the reaction between methylpropene and hydrogen bromide.
 - Draw the displayed structural formula of the product.
 - Name the product of this reaction.
 - Use your answer to (a)(ii) to construct a labelled reaction pathway diagram for the reaction.

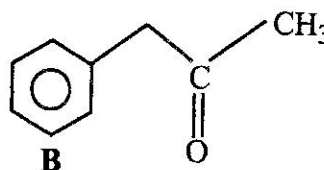
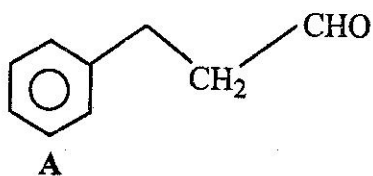
[9]

- (b) The halogenoalkane 2-chlorobutane reacts with ethanolic aqueous sodium hydroxide to produce a mixture of three alkenes with the same molecular formula as methylpropene.
- State the type reaction between 2-chlorobutane and ethanolic $\text{NaOH}_{(\text{aq})}$.
 - Draw the displayed structural formula of any **two** alkenes produced in this reaction.

[3]

[Total: 12]

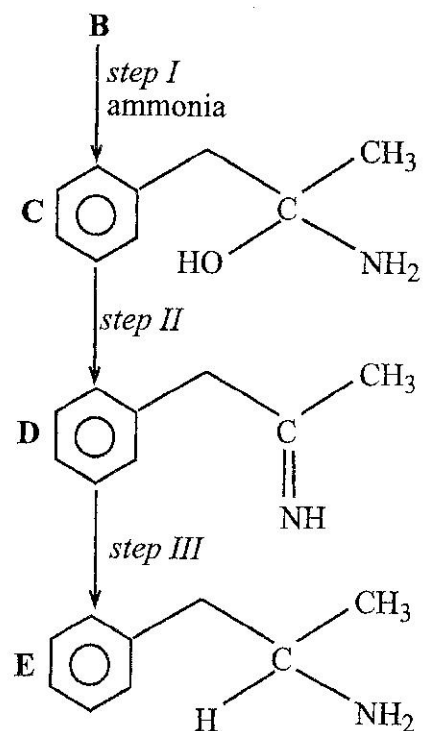
- 7 (a) Compounds A and B are structural isomers:



- Write the molecular formula of A.
- Name the type of isomerism shown by the two compounds.
- Describe a simple test tube reaction that can be used to distinguish A from B.

[5]

- (b) **B** can react with ammonia the same way it does with hydrogen cyanide. The following steps show the conversion of **B** into **E**.



- (i) Suggest the type of reaction in
 1. *step I*
 2. *step II*
 - (ii) Give the reagents and conditions for *step III*.
- (c) Compound **C** can react with ethanoyl chloride in a ratio of 2: 1.
- (i) State the conditions used in this reaction.
 - (ii) Describe the observations made.
 - (iii) Draw the displayed structural formula of the organic product.
 - (iv) Name **one** functional group present in the product.

[3]

[4]

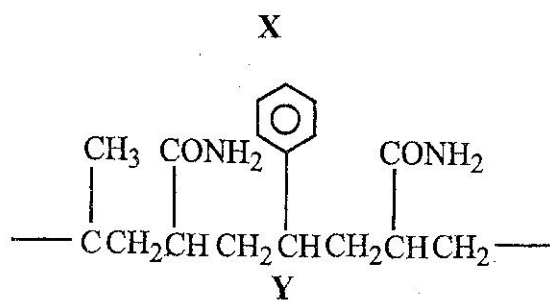
[Total: 12]

8 (a) Define the terms

- (i) *polymer*,
- (ii) *monomer* and
- (iii) *polymerisation*.

[3]

(b) X and Y are segments of synthetic polymers.



(i) Predict the type of polymerisation which produces polymer

- 1. X,
- 2. Y.

(ii) Name the type of linkage in X.

(iii) Deduce the structures of monomers in

- 1. X,
- 2. Y.

[7]

(c) State **two** difficulties associated with the disposal of polymers.

[2]

[Total: 12]